

Problem 1

Problem 1

Problem 1 Which of the following limits are of the form $\frac{0}{0}$?

(a) $\lim_{x \rightarrow 7} \frac{\sin(x-7)}{|x-7|}$

(b) $\lim_{x \rightarrow \frac{\pi}{2}} \frac{\sin(x)}{x}$

(c) $\lim_{x \rightarrow \frac{\pi}{2}} \frac{|\cos(x)|}{x}$

(d) $\lim_{x \rightarrow \frac{\pi}{2}} \frac{\pi - 2x}{\cos(x)}$

(e) $\lim_{x \rightarrow 2} \frac{x^2 - 4}{x - 4}$

(f) $\lim_{x \rightarrow 4} \frac{x^2 - 4}{x - 4}$

(g) $\lim_{x \rightarrow 4} \frac{x^2 - 16}{x - 4}$

(h) $\lim_{x \rightarrow 3} \frac{x - 4}{\sqrt{25 - x^2} - 4}$

(i) $\lim_{x \rightarrow 3} \frac{x - 3}{\sqrt{25 - x^2} - 4}$

Explanation. (a), (d), (g) and (i)